

NOTE FROM DEVELOPER

Welcome to Clean World! These developer docs should hopefully give you a general idea of how to work with the current version of the project (as of May 2026). Included in this file is a summary of what each doc contains and some general tips for working with the project. My apologies for any random jankiness you may come across as you dig into the code; I did my best to clean up as much of it as I could, but there are still some (perhaps many) rough spots. I wrote the majority of the code currently being used, so if you have any questions, feel free to contact me through Professor Akram or Professor Burris. I wish you the best of luck working with Clean World!

- Graydon Bruyn, May 2026

SUMMARY OF THE DOCUMENTATION

The following is a list of descriptions for each document contained in the “Developer_Docs” folder. Most of them describe game systems, but a few are how-to guides or descriptions of the user interface. The how-to guides are in **green**. The UI descriptions are in **orange**.

Audio_System.pdf

Describes how the system that plays in-game audio works and how to use it. Includes instructions for adding new sound effects.

Scripts Involved: *AudioManager.cs, AudioData.cs*

Creating_A_Build.pdf

Details the process of creating a Clean World build and uploading it to itch.io. Also contains tips for fixing things that can go wrong in the build.

Creating_A_New_Tile.pdf

Goes through the process of how to create a new tile, step by step. It includes information on how to use the tile factory and on how to delete tiles.

Scripts Involved (All of these are Editor Scripts): *tileFactoryEditor.cs, tilePrefabEditor.cs, deleteTileButton.cs, SetObjectMaterialsToDitherShader.cs*

Developer_Cheats_System.pdf

Describes how the cheats-system works and can be activated.

Scripts Involved: *creativeModeScript.cs*

Events_System.pdf

Explains how to use the events system and how to add new events.

Scripts Involved: *GameEventManager.cs, EventType.cs*

For_AI_Assistants.pdf

Provides guidelines to any LLMs that may be editing the Clean World project.

Grid_System.pdf

Describes how to use the grid manager, which keeps track of all the tiles that have been placed.

Scripts Involved: *GridManager.cs, GridChunk.cs*

Ground_Chunk_Expansion_System.pdf

Contains information on how to use the system that handles when the player switches between and purchases new areas.

Scripts Involved: *GroundAreaExpansion.cs, TileMeshLoadManager.cs, TileMeshLoader.cs, CameraMove.cs.*

Level_Data_Manager.pdf

Explains how the system that keeps track of the amount of money & carbon in the game works.

Scripts Involved: *LevelManager.cs*

People_Movement_System.pdf

Describes the system that controls the people that wander around the world. Note that the system that keeps track of the number of people & the world's current residential capacity is a separate system called the "People_System".

Scripts Involved: *PeopleMovementManager.cs, MovementPathMap.cs, MovementPathGenerator.cs, PersonWalk.cs*

People_System.pdf

Describes the system that tracks the number of people added by the player, ensures there are enough residences for the people, and determines if there are enough employees for the workplaces. Also has some descriptions of how the People Panel UI works.

Scripts Involved: *PeopleManager.cs, PeoplePanel.cs, PressPeopleButton.cs*

Progression_System.pdf

Contains information on the system that controls game-progression and on how to add new progress events.

Scripts Involved: *ProgressionManager.cs*

Road_Connection_System.pdf

Explains the system that activates buildings when they're connected by roads. This doc also explains the separate but related system that describes how road tiles change their meshes to connect to each other.

Scripts Involved: *RoadAndResidenceConnectionManager.cs, ActivatableTile.cs*

Tile_Placement_System.pdf

Gives an overview of how tiles are placed and how tile overlap/destruction rules work.

Scripts Involved: *BuildingSystem.cs, ObjectDrag.cs*

Tile_Select_Panel.pdf

Goes over how the Tile Select Panel works, which is the panel on the right side of the user's screen that holds the buttons for all the tiles that can be placed.

Scripts Involved: *TileSelectPanel.cs, TileSelectPanelScroll.cs*

Tile_Transparency_System.pdf

Describes the system that makes tiles partially transparent whenever the mouse is hovering over them.

Scripts Involved: *MouseHoverTransparency.cs, MouseHoverHideTile.cs*

Tile_Type_Counter_System.pdf

Contains an explanation of the system that keeps lists of different kinds of tiles that have been placed in the game. (For example, it keeps track of residences, roads, workplaces, etc.)

Scripts Involved: *TileTypeCounter.cs*

Tool_Tip_System.pdf

Explains how tool tips, which appear over tiles' buttons, work. It also has information on how to add new elements to the tool tips.

Scripts Involved: *ToolTipManager.cs, ToolTipFormatting.cs*

Tutorial_Tip_System.pdf

Provides a description of how the tutorial tips work. (Note: the tutorial tips are the yellow hints that appear in the upper left of the screen to help the player.) It also gives a step-by-step explanation of how to add a new tutorial tip.

Scripts Involved: *TutorialTipManager.cs, TutorialTip.cs*

UI_Notifications_System.pdf

Explains the system behind the notifications (primarily error messages) that are displayed to the player.

Scripts Involved: *unableToPlaceTileUI.cs, ErrorGraphicDisplayer.cs*

GENERAL TIPS FOR DEVELOPMENT

Save Local Backups

Don't rely exclusively on GitHub for all your backups. Unless you're a Git wizard (which I'm absolutely not), things can go wrong easily, so I recommend routinely copying the project files and saving them onto a USB.

Make Lots of Test Builds

The game likes to behave differently in builds than it did in the editor. To make sure everything doesn't break in the build, make builds routinely.

Watch Out for Canvas Scaling in the Builds

UI Canvases always scale wrong in the build. To get around this issue, any time I use a constant value for a screen position in my code, I multiply it by "CanvasScalarFactor.CSF.GetScaleFactor()". It usually takes some experimenting to get it to work right.

Use Events Over Direct References

This is easiest to explain with an example. Whenever I want the Place Tile sound effect to play, I don't go into the PlaceTile() function and at the end add a line like SoundManager.PlayPlaceTileSound(). Instead, I call an event like this: GameEventManager.GetEvent(EventType.E.PlaceTile).Invoke(). Then, in the sound manager, I add a listener for that event to play the sound whenever the event is called. In the long run, this kind of programming helps untangle the code a lot.

- Graydon Bruyn, May 2026